

ELECTRIC VEHICLE MARKET ANALYSIS

2020 - 2026

Exploratory Data Analysis and Data Preparation Report

Prepared by

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Dataset: ev_market_2026.csv (2,000 records, 24 variables)

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

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1. Dataset and Objective

The dataset `ev_market_2026.csv` contains 2,000 electric vehicle records covering model years 2020 to 2026. Each row is one vehicle configuration described by 24 columns: identification details (brand, model, year, variant), battery and range figures, performance figures, physical dimensions, and commercial data such as price, annual sales, customer rating and market segment.

The aim of the work was to describe how the market is structured and to prepare the data for modelling. All analysis was done in Python using Pandas, NumPy, Matplotlib, Seaborn and Scikit-learn.

2. Data Quality

The dataset was checked for missing values, duplicates, impossible values and inconsistent category labels. It passed all four checks: zero nulls across all 24 columns, zero duplicate rows, and all numeric values within physically sensible limits (price 16,394 to 269,775 dollars, battery 40 to 119.9 kWh, range 150 to 447 miles, horsepower 150 to 1,000, weight 1,500 to 3,000 kg). The categorical columns returned 20 brands, 5 variants, 3 drive types, 6 body types, 6 countries and 4 market segments, all consistently spelled. No cleaning or imputation was needed.

3. Key Statistics

Table 1. Descriptive statistics for selected variables (n = 2,000)

Variable	Mean	Median	Std. dev.	Range
Price (USD)	78,873.91	73,013.02	35,296.62	16,394 - 269,775
Battery capacity (kWh)	74.66	75.10	20.08	40.0 - 119.9
Range (miles)	261.83	261.00	71.03	150 - 447
Charging speed (kW)	155.21	132.10	80.98	50.0 - 349.9
0-60 mph (seconds)	5.63	5.67	1.42	2.50 - 8.00
Horsepower	665.74	683.00	267.12	150 - 1,000
Weight (kg)	1,886.16	1,740.50	383.77	1,500 - 3,000
Annual sales (units)	129,653.59	-	105,775.15	656 upwards
Customer rating	3.57	-	0.27	3.00 upwards

Table 2. Distribution by market segment

Segment	Vehicles	Share
Premium	712	35.6%
Mid-range	642	32.1%
Luxury	495	24.8%
Budget	151	7.6%

4. Findings

1. Price is strongly right-skewed. The mean of 78,873.91 dollars sits above the median of 73,013.02, and the maximum is sixteen times the minimum. Half of all vehicles fall between 52,326 and 99,780 dollars.
2. The market is concentrated at the upper end. Premium, mid-range and luxury together account for 92.4 per cent of records, while budget vehicles make up only 7.6 per cent. This explains why the average price is so high: the sample itself is weighted towards expensive cars.

3. Battery capacity and range have converged. Both are almost perfectly symmetrical, with mean and median nearly identical. A typical vehicle carries about 75 kWh and delivers about 261 miles, suggesting the industry has settled on a common standard.
4. Charging speed is the real technical divider. Its mean of 155.21 kW is well above its median of 132.10, with a spread from 50 to 349.9 kW. Models differ far more in how fast they charge than in how far they travel.
5. Performance is high throughout the dataset. Average horsepower is 665.74 and even the slowest vehicle reaches 60 mph in 8.00 seconds, reflecting the high torque electric drivetrains provide even outside the luxury segment.
6. Sales volume is highly concentrated. Annual sales have a standard deviation of 105,775 against a mean of 129,654, with a minimum of 656 units, meaning a few models capture most of the volume while many sell in small numbers.
7. Customer satisfaction barely varies. Ratings have a standard deviation of only 0.27 around a mean of 3.57, so buyers report similar satisfaction regardless of price paid.
8. Horsepower and range contain no statistical outliers. Their IQR fences (-302 to 1,684 and 45 to 469) lie outside the observed values entirely. Price and annual sales do have flagged high values, but these are genuine luxury and best-selling models, so they were retained rather than removed.

5. Data Preparation

Six features were engineered from the existing columns: range_efficiency (miles per kWh), price_per_range_mile, power_to_weight, torque_to_weight, battery_to_weight and vehicle_age. These express relationships the raw columns hold only implicitly, most usefully range efficiency, which allows a small car and a large one to be compared fairly.

The categorical columns were then one-hot encoded with drop_first set to true, which avoids the dummy variable trap by treating one category in each column as the baseline. Eleven numeric columns were standardised with StandardScaler so that large-scale variables such as annual sales would not dominate distance-based algorithms.

Final validation confirmed 2,000 rows and 62 columns, with no missing values and no duplicates. No records were lost during processing and the dataset is ready for modelling.

6. Conclusion

The dataset describes a market that has standardised on its core technology while remaining stratified by price. Battery size and range cluster tightly around 75 kWh and 261 miles and performance is strong in every segment, yet prices vary by a factor of sixteen and more than nine vehicles in ten sit at mid-range level or above. Sales are concentrated in a few models while satisfaction stays flat across the price range, which suggests success depends more on brand position than on measurable vehicle quality.

The next steps are a correlation analysis to quantify the relationships described here, grouped comparisons by brand and segment, and a year-on-year trend analysis of price, range and charging speed.